

School of Computer and Systems Sciences
Jawaharlal Nehru University

CS - 403, Data Structures

Time: 1 hr. 30 mins.

MCA, 1st Sessional Exam. 2024

Full marks: 10x3 = 30

Attempt any three (including subparts)

- 1 a) What is the difference between direct and indirect recursion ? What is tail recursion and what is its advantage?
- b) Write a recursive (but not tail-recursive) algorithm (using C pseudocode) to find the factorial of a given number n (where the value of n is provided by the user as input) and specify the number stack frames (of the call stack) required for executing the function implementing this algorithm.
- c) Write an algorithm (using C pseudocode) to store a list of n elements taken from the user as input (where the value of n is taken from the user as input) and perform any of the following operations depending upon user's choice.
 - i) Insert an element at the k^{th} position in the list (where the value of k is taken from the user as input)
 - ii) Sort the list of elements (stored in the array) in ascending order.

(2+3+5)

- 2 a) Write a tail-recursive algorithm (using C pseudocode) which takes the value of n from the user as input and displays the n^{th} term of the Fibonacci series as output. Also specify the number stack frames (of the call stack) required (for the compiler which recognizes tail-recursion and implements it) for executing the function implementing this algorithm.
- b) A complex-valued matrix X is represented by a pair of matrices (A, B) , where A and B contain real values. Write an algorithm (using C pseudocode) that computes the product of two complex-valued matrices (A, B) and (C, D) , where $(A, B) * (C, D) = (A+iB) * (C+iD) = (AC-BD) + i(AD+BC)$. Determine the number of additions and multiplications if the matrices are all $n \times n$.
- c) Explain (by specifying suitable examples of programming languages) 0 - based indexing, 1-based indexing and n -based indexing for arrays.

(3+4+3)

- 3 a) Write down algorithms (using C pseudocode) to perform the following operations on a stack (implemented using array) of size n (where the value of n is provided by the user as input).
 - i) Push an element (provided by the user as input) to the stack only if there is no overflow.
 - ii) Pop k number of elements (provided by the user as input) from the stack (where the value of k is also taken from the user as input) only if it is non-empty.

- b) Write an algorithm (using *C* pseudocode) to perform the addition of two polynomials [where each polynomial is stored the array as a list of elements in the form $(m, e_{m-1}, b_{m-1}, e_{m-2}, b_{m-2}, \dots, e_0, b_0)$ where, the first entry m represents the number of non-zero terms followed by two entries (e_i, b_i) for each non-zero term (representing an exponent-coefficient pair) in the order $e_{m-1} > e_{m-2} > \dots > e_0 > = 0$] and print the resultant polynomial.

(4+6)

- 4 a) Let an array be represented as $A(2:20, 3:30, 1:25, 0:25, 5:30)$. Assuming row major representation of the array (using sequential byte addressing scheme) with 2 bytes per element. If k is the address of $A(4,5,6,7,5)$ then the address of the element represented as $A(5,10,15,6,12)$ is _____ [Fill in the blank correctly].

- b) Write down algorithms (using *C* pseudocode) to perform the following operations on a non-circular queue (implemented using array) of size n (where the value of n is provided by the user as input).

- i) Insert k number of elements (provided by the user as input) to the non-circular queue (where the value of k is also taken from the user as input) only if there is no overflow.
- ii) Delete an element from the queue only if it is non-empty
- iii) Reverse the elements of the non-circular queue using stack

(4+6)

- 5 a) Let a sparse matrix be stored in an array $X(0:t,0:2)$ where t is the number of nonzero terms. The elements $X(0,0)$ and $X(0,1)$ contain the number of rows and columns of the matrix respectively. $X(0,2)$ contains the number of nonzero terms. Let $X(i,0)$, $X(i,1)$ and $X(i,2)$ contain the row number of the element, the column number of the element and the value of the non-zero element in the matrix respectively, for $1 \leq i \leq t$. Write an algorithm (using *C* pseudocode) to store two sparse matrices (taken as input from the user) in the form (as specified above using X), perform addition of these matrices, store the result in the form (as specified above using X) and display the result (in the form which is usually understood by an user unfamiliar with the representation specified using X).

- b) Specify the output of the following recursive function for the call `tower(4, 'A', 'B', 'C');`

```
int tower(int n, char source, char inter, char dest)
{
    if(n>0)
    {
        tower(n-1, source, dest, inter);
        printf("\n Move disk %d from %c to %c", n, source, dest);
        tower(n-1, inter, source, dest);
    }
    return 0;
}
```

(5+5)

School of Computer & Systems Sciences
Jawaharlal Nehru University
Data Structures

CS-403

End-Semester Exam. 2024

Time: 3 hrs.

Full Marks: 10x5 = 50

Attempt any five questions (including subparts)

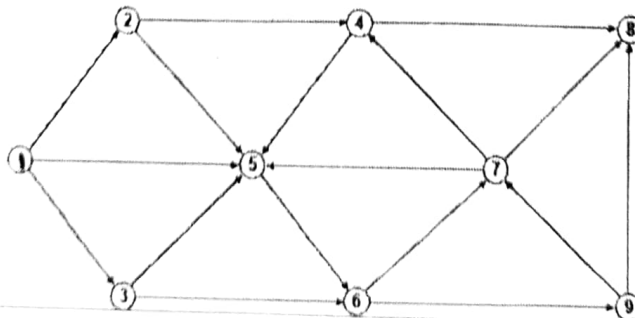
- 1 a) Obtain an addressing formula for the element $A(i_1, i_2, \dots, i_n)$ in an array declared as $A(l_1:u_1, l_2:u_2, \dots, l_n:u_n)$. Assume a sequential column major representation of the array with w bytes per element and α as the address of $A(l_1+1, l_2+2, l_3+3, \dots, l_n+n)$ where $l_i+i \leq u_i$.
- b) Write an algorithm (using C pseudocode) to implement a priority queue using single linked list, where the element having the minimum value is given the maximum priority.
- c) Prove that for any nonempty binary tree, T , if n_o is the number of terminal nodes and n_2 is the number of nodes of degree 2, then prove that $n_o = n_2 + 1$.

(3+4+3)

- 2 a) Write algorithm(s) (using C pseudocode) to insert an element into a circular-queue where the circular-queue is implemented using a doubly linked list. Specify its time complexity using Θ notation.
- b) State whether the following are true or false.
- i) $6 \cdot 2^n + n^2 = O(2^n/6)$ ii) $6 \cdot 2^n + n^2 = \Omega(n^{10000})$
- iii) $10n^2 + 4n + 2 = \Theta(2^n)$ iv) $10n^2 + 4n + 2 = O(n^{2.5})$
- v) $n/500000 = \Omega(500000)$ vi) $500000 = \Theta(5)$
- c) Write algorithms (using C pseudocode) to perform the following:
- i) to convert any given infix expression (taken as input from the user) to its postfix form using stack and display the postfix form obtained.
- ii) to evaluate/calculate the value from the given postfix expression using stack and display the value.

(2+3+5)

- 3 a) Write an algorithm (using C pseudocode) to implement Depth First Traversal of a weighted graph having n number of vertices represented using adjacency matrix. Specify the time complexity of your algorithm.
- b) What is the output of your algorithm for the following graph (in which all the edges have same weights) if:
- i) the traversal starts from vertex 1 ii) the traversal starts from vertex 7



c) Write an algorithm (using C pseudocode) to delete an element from a stack (where the stack is implemented using a single linked list). Specify its time complexity using Θ notation. (3+4+3)

4 a) Specify the postorder sequence for the binary tree whose inorder and preorder sequences are as follows:-

Inorder: B C A E D G H F I

Preorder: A B C D E F G H I

- b) Write an algorithm (using C pseudocode) to print the data present in all the nodes of a binary tree (having n nodes), level by level. Within a level the data (present in the nodes) are to be printed from the left node to the right node. What is the time complexity of your algorithm?
- c) Write recursive algorithms (using C pseudocode) for inorder and postorder traversal of a binary tree (where the tree is implemented using array) and specify their space and time complexities. (4+3+3)

5 a) Write an algorithm (using C pseudocode) to reverse the direction of the links of a singly linked list during traversal.

b) What is the maximum number of items in a B-tree of order m and height h .

c) Construct a B-tree of order 5 with the keys arriving in the following sequence: 3, 7, 9, 23, 45, 1, 5, 15, 30, 27, 13, 10, 8, 19, 4, 31, 36, 55, 2, 6, 11
Then delete the keys in the following sequence: 4, 5, 9, 3, 15 and show the resulting B-tree. (5+1+4)

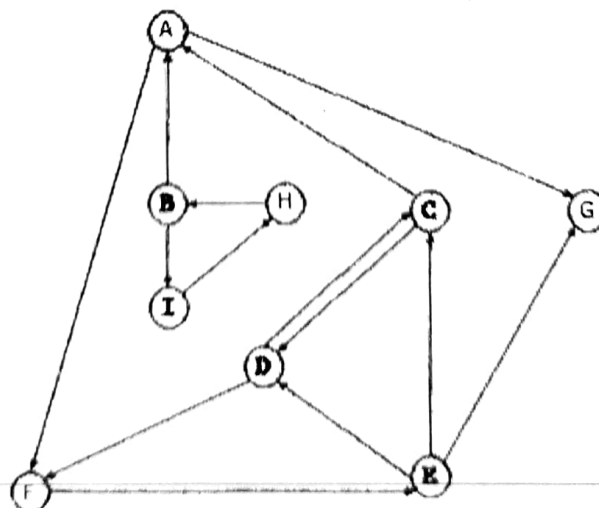
6 a) If $f(n) = a_m n^m + \dots + a_1 n + a_0$ where $a_0, a_1, \dots, a_m$ are constants then prove that $f(n) = \Theta(n^m)$ for $n \geq 1$ and $a_m > 0$.

- b) Write an algorithm (using C pseudocode) to search a key in a Binary Search Tree (BST) and delete it such that the resulting tree is a BST. Can the same algorithm be used for deleting an element in an AVL tree such that the resulting tree is an AVL tree? Specify the worst case time complexity to search a key in a BST and in an AVL tree using Θ notation.
- c) Explain the advantages and disadvantages of using AVL trees. Explain why the time complexity analysis of our algorithm using asymptotic notations may fail when we use disk storage. (3+5+2)

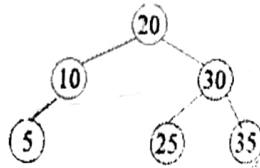
7 a) What is the output for Breadth First Traversal of the following graph if:

i) the traversal starts from vertex A

ii) the traversal starts from vertex I

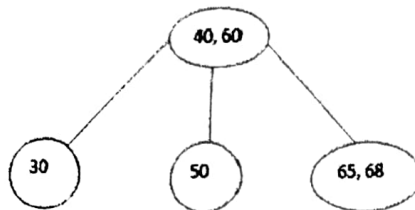


- b) Write non-recursive algorithms (using C pseudocode) for the inorder and preorder traversal of a binary tree having n nodes and specify their best case and worst case time complexities using Θ notation.
- c) Insert the keys in the tree (given below) arriving in the following sequence: 40, 45, 34 and then delete key 20 from it such that the resulting tree is an AVL tree.

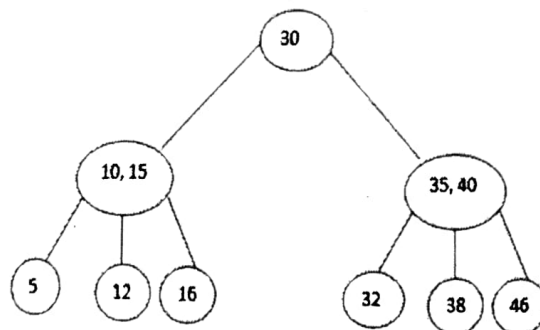


(3+5+2)

- 8 a) Insert the key 62 into the following 2-3 tree such that the resulting tree is a 2-3 tree.



- b) From the following B-tree of order 3 delete the key 30 such that the resulting tree is a B-tree



- c) Write the algorithm (using C pseudocode) for Breadth First Traversal of a graph (containing n number of vertices and e number of edges) represented using adjacency list. What is the time complexity of your algorithm?

(2+2+6)

- 9 a) Write two algorithms (using pseudocode), one should be a non-recursive algorithm for inorder traversal of threaded binary tree and the other should be an algorithm for inorder traversal of non-threaded binary tree. Compare and analyze their space and time complexities.
- b) Prove that the maximum number of nodes on level i of a binary tree is 2^{i-1} for $i \geq 1$ and the maximum number of nodes in a binary tree of depth k is $2^k - 1$ for $k \geq 1$.

- c) For the following algorithm (*tower*), specify its time and space complexity using Θ notation

Algorithm *tower*(*n*, *source*, *inter*, *dest*)

```
{
    if(n>0)
    {
        tower(n-1, source, dest, inter);
        print("Move disk", n, "from", source, "to", dest);
        tower(n-1, inter, source, dest);
    }
}
```

(6+2+2)

- 10 a) Write an algorithm which swaps the left and right children of every node of a binary tree represented using linked representation.
 b) Write algorithms (using C pseudocode) to delete an element from a Binary Search Tree (having *n* nodes) and specify its best case and worst case time complexity using Θ notation.
 c) For the following function [where, *end_index* - *start_index* = *n*-1], the worst case time complexity of **foo** (using Θ notation) is _____ and the best case time complexity (using Θ notation) of **foo** is _____ and the worst case recursion stack space required by **foo** (using Θ notation) is _____

```
int foo(int array[], int start_index, int end_index, int element){
    if (end_index >= start_index){
        int middle = start_index + (end_index - start_index) / 2;
        if (array[middle] == element)
            return middle;
        if (array[middle] > element)
            return foo(array, start_index, middle-1, element);
        return foo(array, middle+1, end_index, element);
    }
    return -1;
}
```

(4+4.5+1.5)
